Lesson 5
Cash Flow Analysis

Excel in Business

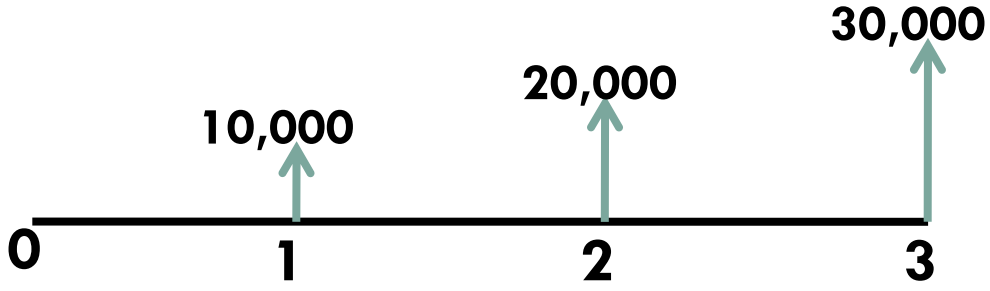
Question 1

Consider the following cash flow:

Year	Payment @ end of year
1	10,000
2	20,000
3	30,000

Would you pay Rs 55,000 today to get this sequence of cash flow?

Question 1(cont..)



Suppose the prevailing market rate is 10% compounded annually.

Then the NPV of the payments is

$$\frac{10000}{1.1}+\frac{20000}{1.1^2}+\frac{30000}{1.1^3}=48{,}159.28$$

NPV of future earnings

Question 1(cont..)

If we paid Rs 55,000 today for them, the NPV for the entire investment would be

$$\text{Rs } -55{,}000 + 48{,}159 = -6841,$$

a loss of Rs 6841 in today's terms.

On the other hand, if you are offered the same future cash flows for Rs 45,000 then

$$\text{NPV} = \text{Rs } -45{,}000 + 48{,}159 = \text{Rs } 3159,$$

a profit of Rs 3,159 in today's terms.

Application of NPV: Comparing two projects

Which is a better investment plan?

Year	Project A (in millions)	Project B (in millions)
0	-800	-800
1	600	400
2	400	400
3	200	400

Application of NPV: Comparing two projects(cont..)

Discount rate =10%

$$NPV(A) = -800 + \frac{600}{1.1} + \frac{400}{1.1^2} + \frac{200}{1.1^3} = 226.30$$

$$NPV(B) = -800 + \frac{400}{1.1} + \frac{400}{1.1^2} + \frac{400}{1.1^3} = 194.74$$

Since NPV(A) > NPV(B), project A is a better investment than project B.

To summarize:

- NPV is used to make investment decisions
- A positive NPV represents profit, and
- a negative NPV represents loss (compared to letting your money earn interest).

NPV Function

- NPV

  Syntax:

  = NPV (rate per period, cash flow range)

The NPV function

- Handles only end of the period payments
- Does not include the initial investment since it considers all payments end of the first period onwards

Move to worksheet *NPV, IRR*

XNPV Function

- XNPV

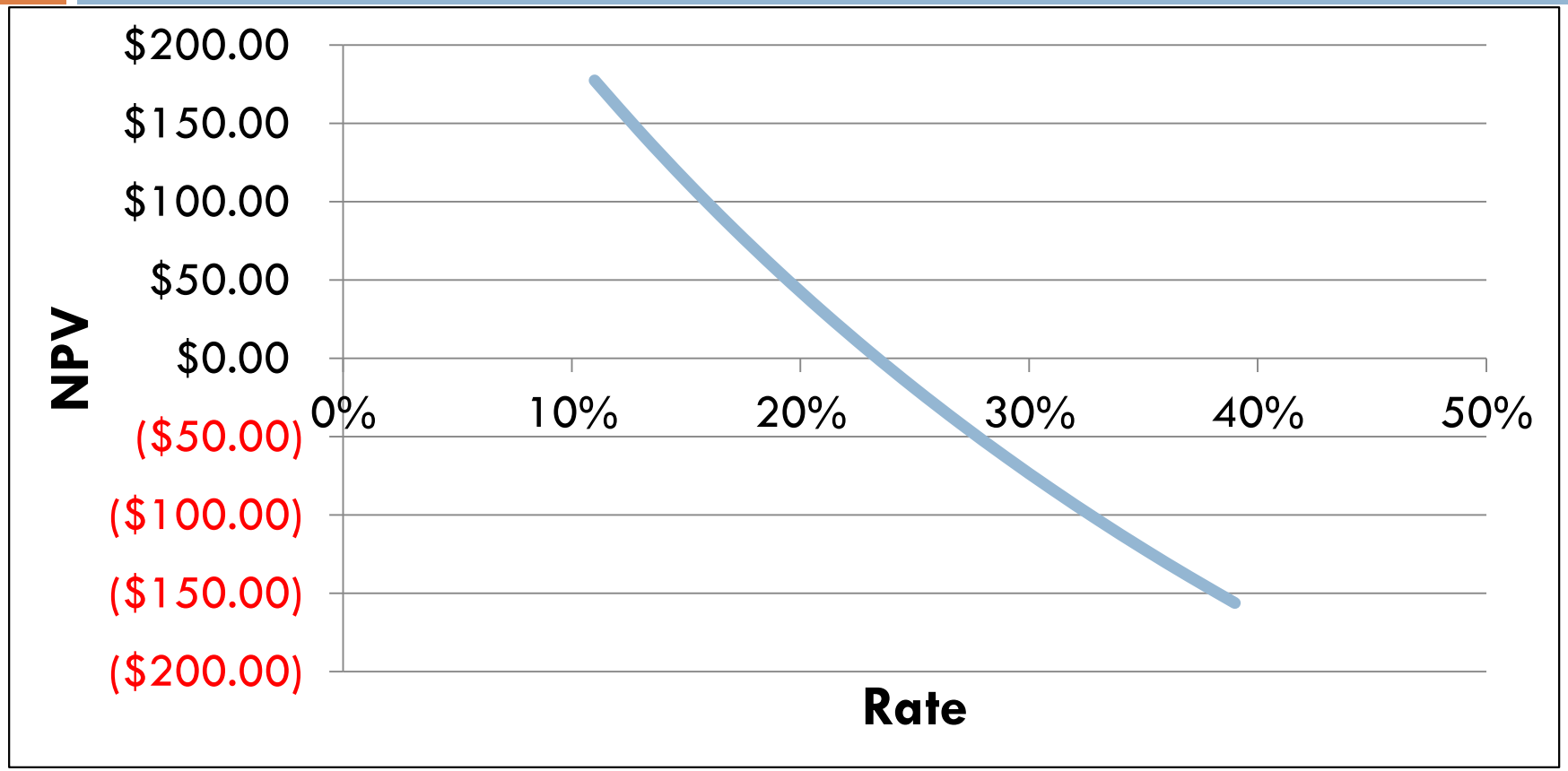
  Syntax:

  = XNPV (annual rate, cash flow range, date)

The XNPV function handles the situation when payments are made at unequal time periods.

Move to worksheet *XNPV, XIRR*

NPV vs Rate



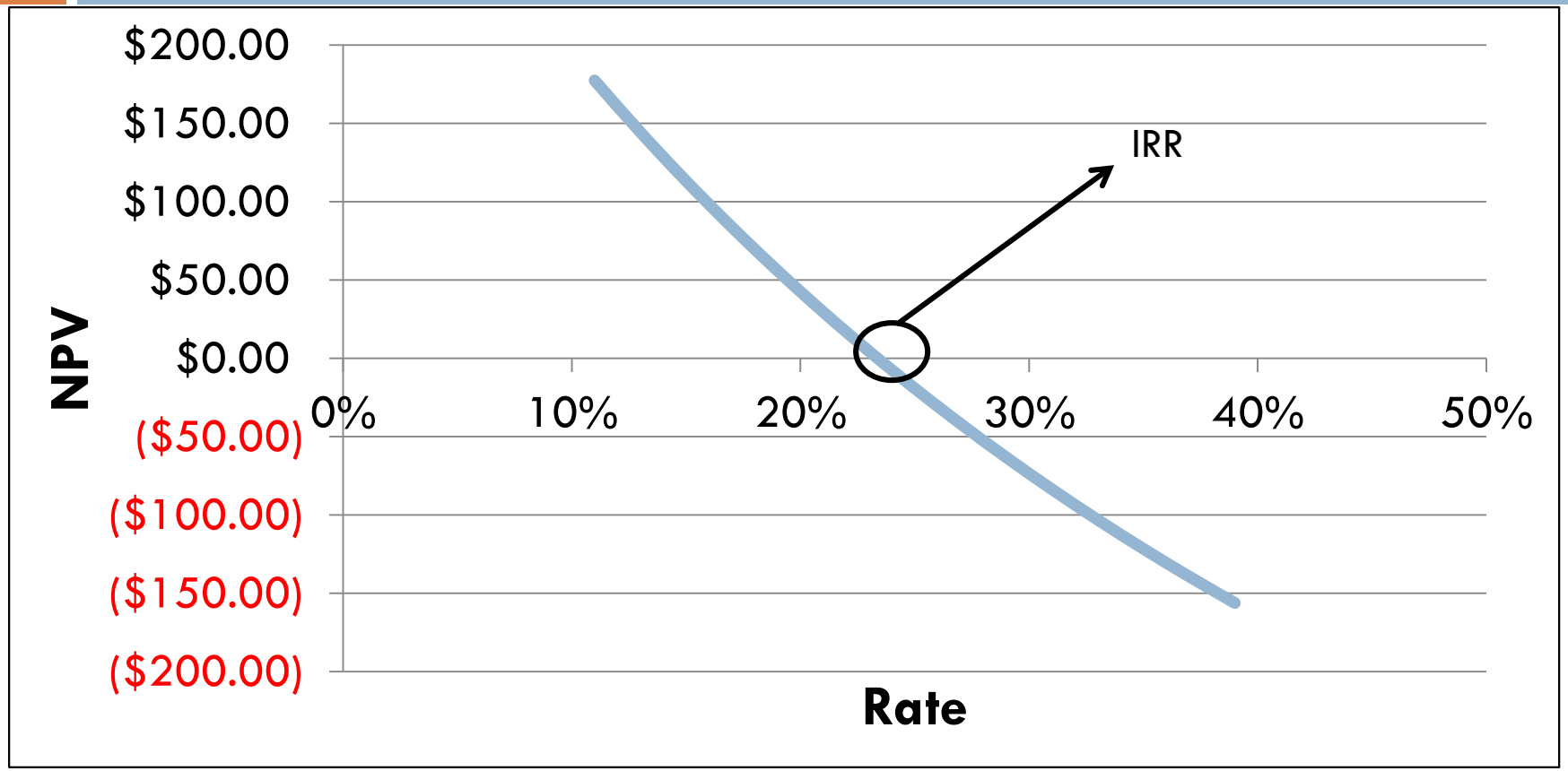
Project B (last example)

Internal Rate of Return (IRR)

The internal rate of return (or IRR) of a cash flow is defined as that annual rate of interest which makes its NPV equal to zero.

IRR is viewed as measuring the annual rate of growth of the investment.

IRR



Project B (last example)

IRR Function

SYNTAX:

= IRR (cash flow range)

- Note that for using this function we must include the initial investment also.

Move to worksheet *NPV, IRR*

XIRR Function

SYNTAX:

= XIRR (cash flow range, dates)

- Note that for using this function we must include the initial investment also.

Move to worksheet *XNPV, XIRR*

IRR: A closer look

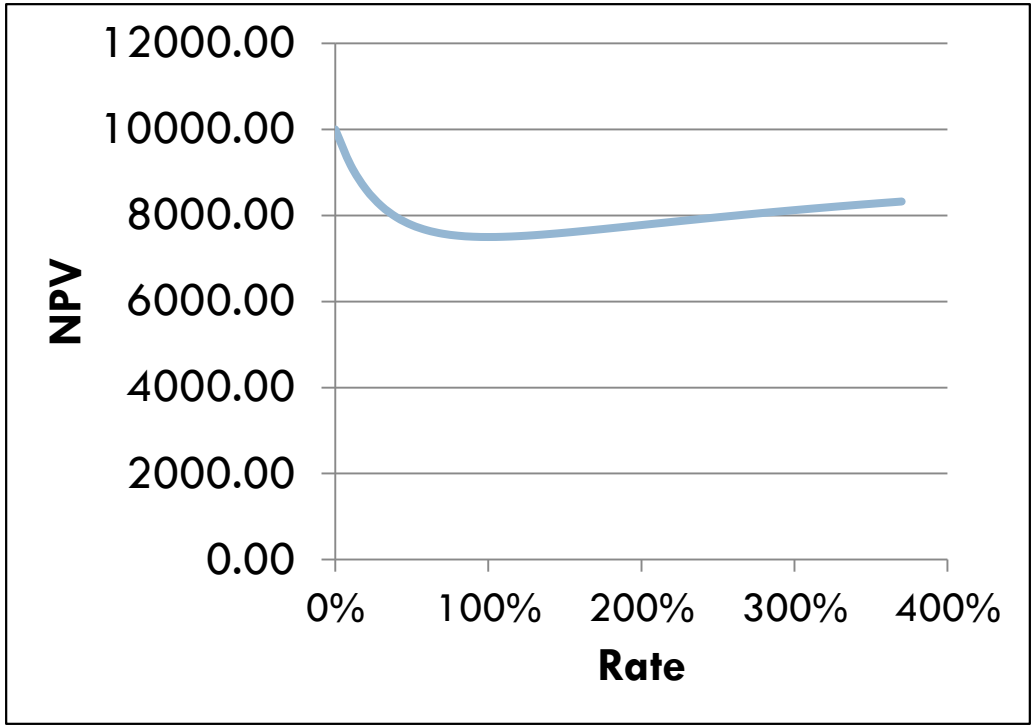
Two possible sources of trouble become visible:

1. What if there is no rate of interest which makes the NPV zero?

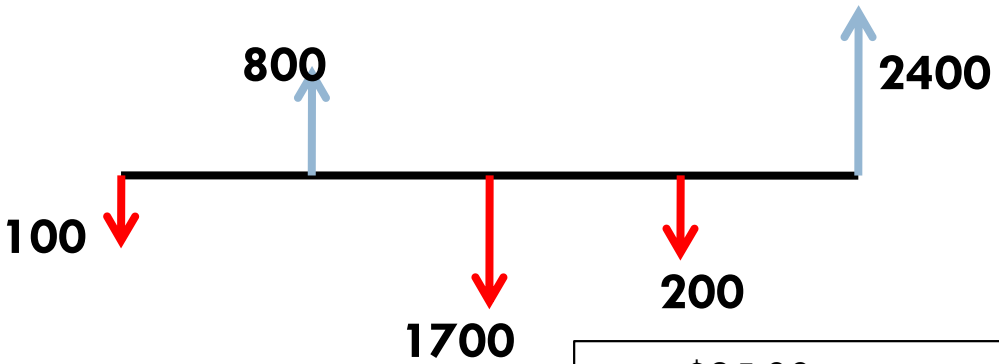
2. What if there are several rates of interest which make the NPV zero?

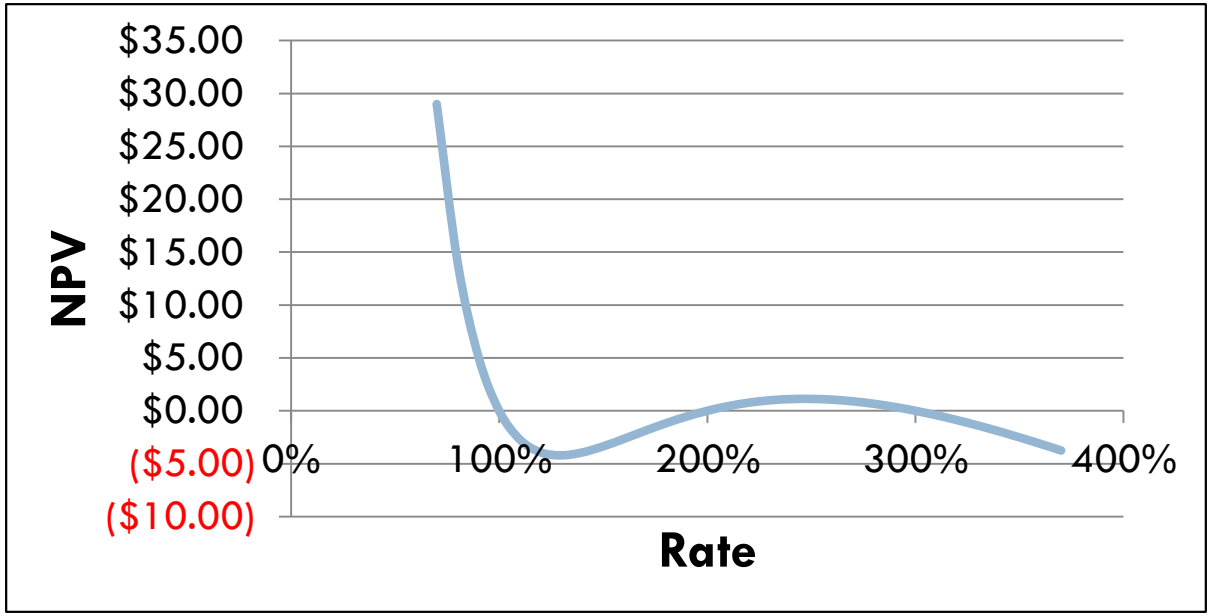
Example: No IRR exists

Year	Cash Flow
0	10000
1	-10000
2	10000



Example: Multiple IRR exists





IRR: A closer look

The situation, where there are several solutions for IRR, is particularly common and happens when a cash flow has many intermingled positive and negative entries.

In such cases, none of the solutions can be preferred over the others and the only honest
description is to say that there is no IRR.

IRR: A closer look

There is only one situation when IRR can be safely defined. This is when the first few entries have one sign, and the remaining have the opposite.

For example: −100, −50, −45, 30, 0, 100, 400.

NPV vs IRR

NPV	IRR
NPV is defined for every cash flow.	IRR is not always defined for every cash flow.

NPV vs IRR

NPV	IRR
It is additive. The NPV of a complicated project can be obtained by summing up the individual NPVs of its parts. This eases book-keeping.	IRR is not additive

NPV vs IRR

NPV	IRR
To calculate NPV one needs market information i.e. market rates.	To calculate IRR, no market information is needed.

NPV vs IRR

NPV	IRR
It gives an answer in terms of absolute profit and loss. No sense of $\frac{\text{Profit}}{\text{Investment}}$	It does not gives any sense of absolute profit but does try to estimate $\frac{\text{Profit}}{\text{Investment}}$

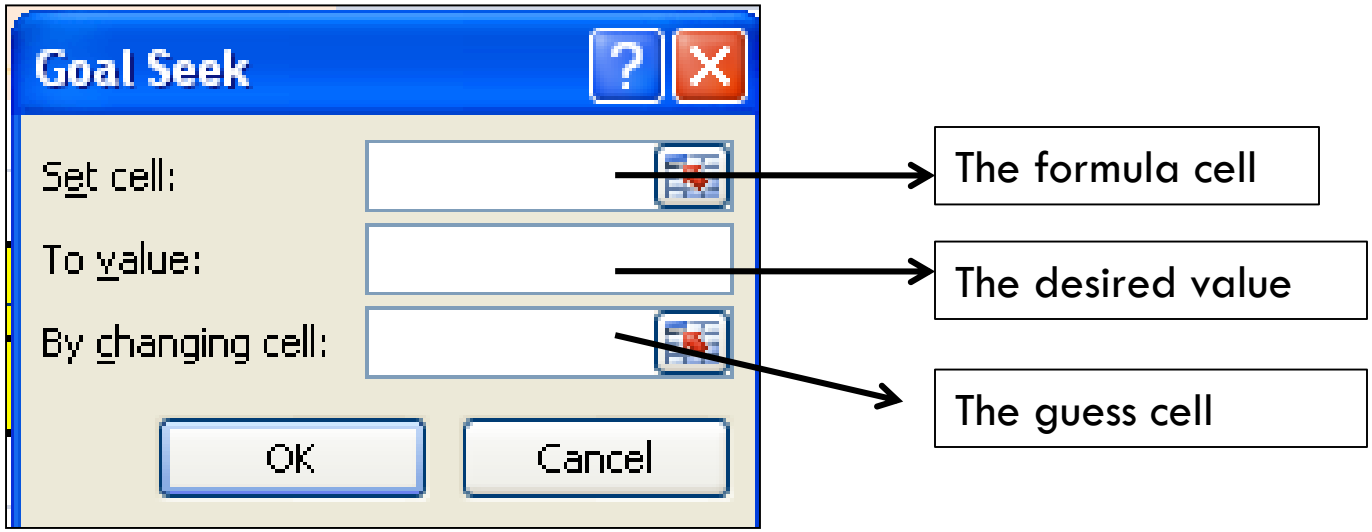
Goal Seek in Excel

If you know the result that you want from a formula, but not the input value the formula needs to get that result, you can use the Goal Seek feature.

For e.g. if one wants to calculate the value of x s.t. $x^2 + 3x+2 = 0$, we can use Goal seek tool of Excel.

Goal Seek in Excel

Data → What If Analysis → Goal Seek



Move to worksheet *Goal Seek*